ModelIt!

Teacher's Guide

# Why Are You Taller Than Your Grandparents?

Modeling Gene Expression, Nutrition, and the Secular Trend in Human Height

## Quick Reference

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|---|---|
| **Grade Level** | Grade 10L1 |
| **NGSS Standard** | HS-LS3-2, HS-LS3-3 |
| **Time** | 45–60 minutes (core lesson) |
| **Materials** | Student Activity Pack, devices with ModelIt! access |
| **Prep** | Print activity packs, test ModelIt! on student devices |
| **Grouping** | Partners or small groups (2–4) |

ModelIt! — Discovery Collective

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# 1 NGSS Standards & CAST Alignment

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**Performance Expectation: HS-LS3-2, HS-LS3-3**

Make and defend a claim based on evidence that inheritable genetic variations may result from new genetic combinations through meiosis, viable errors occurring during replication, and/or mutations caused by environmental factors. Apply concepts of statistics and probability to explain the variation and distribution of expressed traits in a population.

## Three-Dimensional Learning

| | |
|---|---|
| **Science Practice** | **Engaging in Argument from Evidence** — Students make and defend claims about the relative contributions of genetic variation and environmental factors to expressed traits in populations. |
| **Disciplinary Core Idea** | **LS3.B Variation of Traits** — Environmental factors also affect expression of traits. Genes may be turned on or off by environmental conditions, and the expression of genes influences an organism's traits. |
| **Crosscutting Concept** | **Cause and Effect** — Students analyze the causal relationships between nutrition, gene expression, growth hormone release, and final height, distinguishing genetic causes from environmental ones. |

## CAST Testing Alignment

**MULTIPLE CHOICE:**

A study finds that identical twins raised in different countries with different nutrition levels differ in adult height by 8 cm. What does this most strongly support about the determinants of height?

**CONSTRUCTED RESPONSE:**

Using your model, explain why average human height in developed nations increased by approximately 10 cm over the past 150 years despite no significant genetic changes in the population. Include the roles of gene expression, nutrition, and growth hormones in your explanation.

# 2 Scoring Rubric

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### Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

### Scoring Guidance

**Expert (4):** Student demonstrates deep understanding and can transfer knowledge to new contexts.
**Proficient (3):** Student meets grade-level expectations and shows solid understanding.
**Developing (2):** Student is making progress but needs additional scaffolding or practice.
**Beginning (1):** Student needs significant support; consider using Differentiation strategies (Section 7).

# 3 Background Content Knowledge

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**For Your Confidence**

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Over the past 150 years, the average height of people in developed countries has increased by about 10 centimeters — roughly 4 inches. This 'secular trend' happened far too quickly to be genetic evolution (which takes thousands of years). The explanation lies in the remarkable interaction between genes and environment: your DNA sets the blueprint, but the materials and conditions during construction determine the final product.

## The Genetics of Height

Height is a polygenic trait — controlled by over 700 identified genetic variants, each contributing a small amount. Together, genetics accounts for about 80% of height variation within a population. However, this 80% is the variation between individuals in the SAME environment. When environments differ dramatically (as between your generation and your grandparents'), environmental factors can shift the entire distribution by centimeters.

## The Secular Trend in Height

In the Netherlands, average male height increased from 165 cm in 1860 to 183 cm in 2020 — an increase of 18 cm in just 160 years (about 6 generations). Japan saw a 15 cm increase in the 50 years after World War II when nutrition rapidly improved. In contrast, North Koreans are now about 8 cm shorter than South Koreans on average — same genetic background, dramatically different nutrition.

## Nutrition and Gene Expression

Nutrition affects height through multiple pathways. Protein provides amino acids for growth hormone and bone matrix. Calcium and phosphorus form hydroxyapatite crystals in bone. Vitamin D enables calcium absorption. Zinc is essential for growth plate function. When these nutrients are abundant, growth-related genes are fully expressed. When they're scarce, epigenetic mechanisms dial down gene expression — the genetic potential exists but isn't realized.

## Growth Plates and the Window of Opportunity

Growth plates (epiphyseal plates) are zones of cartilage near the ends of long bones. Chondrocytes in these plates divide, producing new cartilage that gradually calcifies into bone, lengthening the bone. Growth hormone and sex hormones (estrogen, testosterone) regulate this process. In late adolescence, rising sex hormone levels trigger growth plate closure (fusion). Once closed, no further height increase is possible — making childhood and adolescent nutrition critically important.

# 4 The LEVER Framework

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### About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

**L** **Locate the System**
Students identify gene expression level, growth hormone release, nutrition quality, bone growth rate, and final height as the key components of the human growth system.

**E** **Establish Relationships**
Students determine that nutrition quality drives gene expression and growth hormone release, which together control bone growth rate, and that the cumulative effect of bone growth determines final height.

**V** **Visualize & Model**
Students build a computational model showing how environmental nutrition interacts with genetic potential through gene expression to determine growth trajectory and final height.

**E** **Evaluate Outcomes**
Students run optimal nutrition, deprivation, and generational comparison scenarios to quantify the environmental contribution to height and model the secular trend.

**R** **Revise & Extend**
Students add sleep quality, physical activity level, or stress hormone levels to model more realistic growth dynamics.

## Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

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| **At the start of a unit** | Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it. |
| **After a reading or lab** | Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes. |
| **As a “what-if” tool** | When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back. |
| **For peer discussion** | Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence. |
| **As a presentation tool** | Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time. |
| **At the end of a unit** | Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved. |

# 5 Slide-by-Slide Facilitation

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## Slide 1: Cover

2 min

**Say:** *"Look at photos of your grandparents as teenagers. Many of you are 3-5 inches taller than they were. Same family. Same genes. So what changed?"*

**Watch For**

Show historical photos of teenagers from the 1940s vs. today. Ask students to observe the visible height difference.

## Slide 2: Learning Objectives

3 min

**Say:** *"Today we're solving a genetic mystery — how can your DNA be basically the same as your grandparents' but you're significantly taller?"*

**Watch For**

Have students read objectives. Pre-teach 'gene expression' and 'epigenetics' — genes are like a library, but environment decides which books get read.

## Slide 3: Big Question

3 min

**Say:** *"If height is 80% genetic and your DNA came from your grandparents, why aren't you the same height?"*

**Watch For**

Quick survey: Are you taller than your grandparents were at your age? Taller than your parents? Discuss patterns.

## Slide 4: LEVER Framework

2 min

**Say:** *"We're modeling the interaction between your genes and your environment to solve this height mystery."*

**Watch For**

Preview LEVER steps. Emphasize this is about gene-ENVIRONMENT interaction, not just genetics.

## Slide 5: Activity 1: Components

8 min

**Say:** *"What factors determine how tall you grow? Which ones come from outside your body?"*

**Watch For**

Guide sorting. Nutrition is the key external input. Discuss why gene expression responds to environment.

## Slide 6: Activity 2: Connections

8 min

**Say:** *"When nutrition improves, what chain reaction happens inside your body? Follow the pathway from food to bone growth."*

**Watch For**

Map the cascade: nutrition → gene expression → growth hormone → bone growth → height. Discuss where the pathway can be limited.

## Slide 7: Activity 3: Simulation

10 min

**Say:** *"Let's model the height difference between your grandparents' nutrition and yours. How many centimeters does nutrition add?"*

**Watch For**

Students predict the nutrition effect. Run generational comparison. Compare to real secular trend data.

## Slide 8: Discoveries

5 min

**Say:** *"Your genes set the ceiling. Your nutrition determines how close you get to it. Your grandparents had the same ceiling — they just couldn't reach it."*

**Watch For**

Show North Korea vs. South Korea height comparison. Same genes, 8 cm difference. Discuss implications.

## Slide 9: STEM Challenge

12 min

**Say:** *"40% of students in this school district are food insecure. If nutrition determines how close kids get to their genetic potential, what should we do about it?"*

**Watch For**

Groups design school nutrition programs that optimize growth. Must include specific nutrients, delivery systems, and evaluation metrics.

# 6 Answer Key & Expected Responses

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## Component Sorting

- **EXTERNAL:** Nutrition Quality
- **INTERNAL:** Gene Expression Level, Growth Hormone Release, Bone Growth Rate, Final Height

## Relationships

- **Nutrition Quality to Gene Expression Level:** POSITIVE (+) — Better nutrition — especially protein, calcium, zinc, and vitamin D — provides the molecular signals that activate height-related genes through epigenetic mechanisms. Nutrient abundance turns on growth genes that remain dormant under scarcity.
- **Gene Expression Level to Growth Hormone Release:** POSITIVE (+) — When growth-related genes are actively expressed, they produce proteins that stimulate the pituitary gland to release more growth hormone. Gene expression sets the biological stage for the hormonal cascade that drives growth.
- **Growth Hormone Release to Bone Growth Rate:** POSITIVE (+) — Growth hormone directly stimulates chondrocyte proliferation in growth plates and promotes IGF-1 production, which drives bone lengthening. More growth hormone during the active growth period means faster bone elongation and ultimately greater height.

## Key Discoveries

- Genetics sets a RANGE of possible heights (about 80% of variation), but environmental factors — especially nutrition — determine where within that range an individual falls
- The secular trend in height is almost entirely explained by improvements in childhood nutrition, not by genetic changes (evolution works too slowly for this)
- Nutrition quality affects gene expression epigenetically — better nutrition activates more growth-related genes and increases growth hormone release
- Growth plates have a limited window of activity — once they fuse in late adolescence, no amount of nutrition or hormones can increase height, making childhood nutrition critically important

# 7 Collaborative Learning Protocols

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### Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

### Model Gallery Walk

**When:** After students build models (LEVER Phase V)
**Setup:** Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
**Protocol:** Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
**Why it works:** Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

### Scientist Fishbowl

**When:** During simulation discussion (LEVER Phase E)
**Setup:** 4-5 students sit in center circle. Rest of class observes with clipboards.
**Protocol:** Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
**Why it works:** Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

### Component Jigsaw

**When:** During component analysis (LEVER Phase L)
**Setup:** Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
**Protocol:** Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
**Why it works:** Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

### Prediction Debate

**When:** Before running simulations (LEVER Phase E)
**Setup:** Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
**Protocol:** Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
**Why it works:** Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

### Pair Programming for Models (Digital Collaboration)

**When:** Throughout model building
**Setup:** Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
**Protocol:** Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
**Why it works:** Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

### Quick Protocol Selection Guide

| | |
|---|---|
| **Short on time (5 min)** | Prediction Debate — gets bodies moving, generates energy fast |
| **Full period available** | Component Jigsaw — deep collaboration, every student contributes |

| | |
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| **After simulation day** | Model Gallery Walk — peer feedback, multiple perspectives |
| **Advanced discussion** | Scientist Fishbowl — builds academic discourse skills |
| **Every day default** | Pair Programming — transforms individual work into dialogue |

# 8 Differentiation & Extensions

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### Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

### Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

### English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

### Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

# 9 Common Misconceptions

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**Why This Matters**

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

| STUDENTS MAY THINK | SCIENTIFIC UNDERSTANDING |
| --- | --- |
| "Height is entirely determined by genetics" | While genetics accounts for about 80% of height variation WITHIN a population sharing the same environment, environmental factors — especially childhood nutrition — can shift entire populations by 10+ centimeters. The 80% figure describes variation between individuals in similar conditions, not the total potential range. North and South Koreans share nearly identical genetics but differ by about 8 cm in average height due to nutrition differences. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "You can increase your height by eating well as an adult" | Once growth plates close (typically ages 16-18 for girls and 18-21 for boys), no amount of nutrition, exercise, or supplements can increase height. Growth plates fuse permanently, and the window for height increase is gone. This is why childhood and adolescent nutrition is so critically important — the window is limited and cannot be reopened. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "Modern humans are evolving to be taller" | The height increase over the past 150 years is too fast for genetic evolution, which requires hundreds to thousands of generations. The secular trend is almost entirely environmental — better nutrition, reduced childhood disease, and improved healthcare allow people to reach more of their genetic potential. In fact, there's no evidence that the genetic potential for height has changed at all. |